Project title

A comparative study of RAG techniques for generating image captions

1. Background and motivation

Automatic image caption generation (image captioning) is a fundamental task in computer vision and natural language processing. Recent models (Vision-Language Models, VLM) produce fluent captions but may:

• omit objects that are present,

• invent objects that are not present (hallucination),

• lack precision regarding attributes, relationships and contexts.

Retrieval-Augmented Generation (RAG) techniques enable the generation process to be grounded in external knowledge (similar images, reference captions, concept graphs) in order to improve the accuracy, relevance and visual grounding of the generated captions.

This project aims to investigate how different RAG techniques influence the quality of generated captions on a subset of the MS COCO dataset. This study aims to produce a reproducible scientific contribution with rigorous statistical evaluation and ablation analysis.

2. Learning objectives

By the end of the project, each group should be able to:

1. Understand and implement a specific RAG technique.
2. Build a complete multimodal pipeline:
   image → visual analysis → retrieval → caption generation.
3. Quantitatively and qualitatively evaluate the captions produced.
4. Analyse errors and hallucinations.
5. Present and discuss the results in a scientific manner.
6. Apply appropriate statistical tests to compare the methods.

7. Conduct an ablation study to identify the contribution of each component.

3. Research question

Main question:

How do different RAG techniques affect the quality, relevance and visual grounding of captions generated on a subset of the COCO image dataset?

Sub-questions:

- Which RAG techniques reduce hallucinations the most?
- What is the trade-off between computational cost and quality gains?
- Do advanced RAG techniques (reranking, self-RAG, corrective RAG) provide a measurable improvement over a naive RAG?
- Are the differences observed between methods statistically significant?
- Which components of the RAG pipeline contribute most to the final performance?

4. Dataset

4.1 Common dataset

1) COCO train2017 (images + captions)

- Total number of images in the knowledge base: 10,000 for training + 100 different images for validation.
- Total number of captions in the knowledge base: 50,000 (5 captions per image).

2) MS COCO val2017 — a stratified subset of 100 images.
- The subset is identical for all groups .
- It is stratified to cover: simple scenes, complex scenes, multiple objects, interactions and varied contexts.
- Total number of test images: 100
- Total number of ground truth captions: 500 (5 captions per image)

4.2 Data splitting

Usage	Source	Size
RAG knowledge base (training)	COCO train2017 (images + captions)	10,000 images
Testing+ evaluation	COCO val2017 (subset)	100 images
Hyperparameter tuning	Validation subset	100 images

Anti-leakage rule: the reference captions for evaluation images must never be used in the knowledge base or retrieved for the same image.

4.3 Statistical exploration of the dataset

A comprehensive statistical exploration of the dataset must be carried out and documented in the report; it must include:

Distribution of object classes: frequency of objects (top 50) via COCO annotations.

Scene complexity: average number of objects per image, distribution of spatial relationships.

Textual statistics: average caption length (number of words/tokens), vocabulary size, and part-of-speech (POS) distribution.

Diversity analysis: study of attributes (colour, shapes) and contexts (indoor/outdoor, day/night)

Daa visualisation: histograms, scatter plots, co-occurrence matrices, etc.

4.4 Preprocessing phase (common to all groups)

4.4.1 Objective The preprocessing phase transforms the raw COCO data into a clean, reproducible, and shared set of artifacts that will be used by all groups.

4.4.2 Inputs

- COCO train2017 images and captions (10,000 images for the knowledge base)+ 100 validation images.
- COCO val2017 stratified subset (1,000 test images).
- COCO annotations (objects, attributes where available, relationships/context metadata).
- Assigned image/text encoders and optional object detectors.

4.4.3 Preprocessing steps

1. Data audit and validation

- Verify image counts: 10,000 training/knowledge-base images, 100 validation images, 1,000 test images.

- Verify 5 reference captions per image where applicable.

- Detect missing, corrupt, duplicated, or unreadable images.

- Check image–caption integrity and remove invalid samples only if documented.

2. Split creation and anti-leakage control

- Create and freeze three splits: knowledge base (training), validation/tuning (100 images), test/evaluation (1,00 images).
- Ensure that test-image reference captions are never included in the knowledge base.
- Ensure that a test image is never retrieved for itself.
- Store split IDs in versioned files and document the anti-leakage audit.

3. Image preprocessing

- Convert images to RGB and correct EXIF orientation.
- Resize/normalise according to the assigned image encoder requirements.
- Preserve aspect ratio where required.
- Optionally run object detection to extract visual metadata.

4. Caption preprocessing

- Keep the original reference captions unchanged for evaluation.
- For retrieval/indexing only, apply documented cleaning: tokenisation, lowercasing where appropriate, whitespace normalisation, and optional removal of irrelevant punctuation.
- Store both raw and cleaned captions with clear field names to avoid evaluation leakage.

5. Metadata extraction

- Extract detected objects, attributes, relationships, and scene/context metadata using the permitted detectors or annotation fields.
- Store metadata in JSON/CSV with image IDs.
- Document detector versions and confidence thresholds.

6. Embedding generation

- Generate image embeddings using the selected image encoder (CLIP, SigLIP, etc.).
- Generate text embeddings using Sentence-BERT or the CLIP text encoder.

- Record embedding dimensions, batch size, normalisation, and model versions.
- Save embeddings in a reproducible format (e.g:npy).

7. Knowledge-base construction

- Build the vector database/index (FAISS, ChromaDB, or LanceDB).
- Document index type (flat, IVF, HNSW, etc.), similarity metric (cosine, L2, inner product), database size, and indexing time.
- Link each vector to its image ID, captions, and metadata.

8. Quality control

- Check for missing embeddings, NaNs, duplicate IDs, and caption mismatches.
- Run a leakage audit.
- Produce sample visualisations and statistics: object-class frequency, scene complexity, caption length, vocabulary size, POS distribution, and diversity analysis.
- Validate that the frozen release works with the retrieval pipeline.

4.4.4 Validation and freezing

The validation subset of 100 images is used to select the best preprocessing configuration. Once selected, the preprocessing output is frozen and shared with all groups. No group may modify the frozen test split or use test reference captions during retrieval or generation.

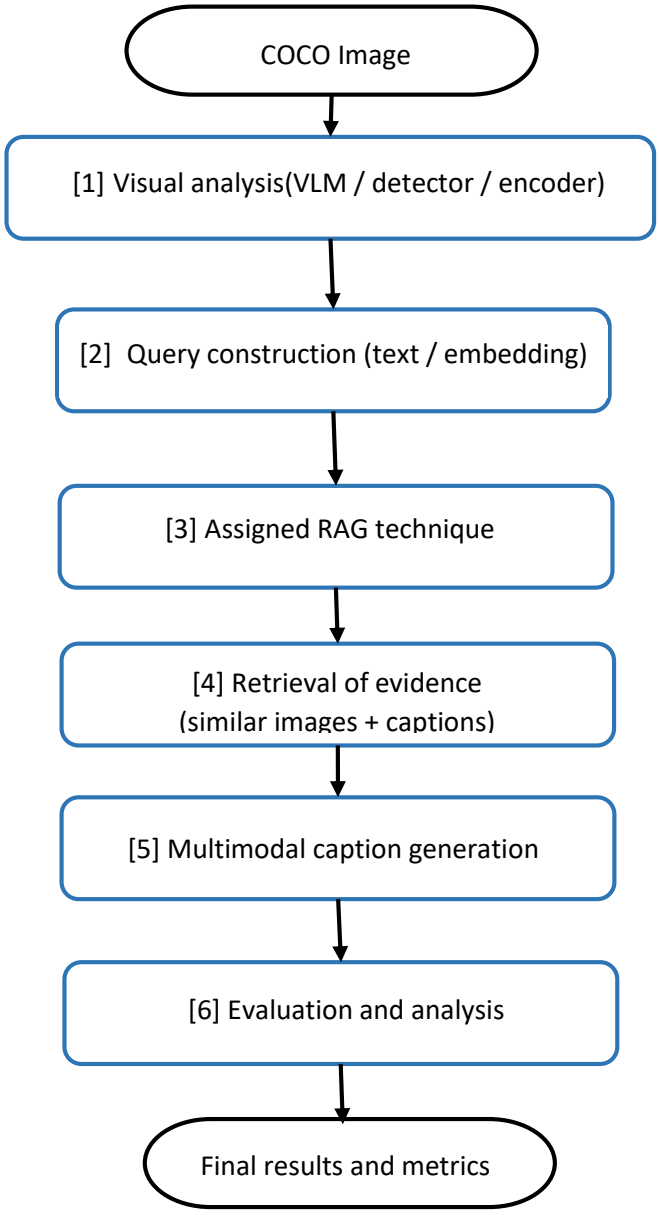
5. Assigned RAG techniques

The scope has been narrowed to 10 RAG techniques, with each technique assigned to three groups to enable within-technique comparisons. All groups will use the same dataset, consisting of the best-performing output from the preprocessing stage, to ensure a fair and consistent comparison across techniques.

Group	RAG technique	Brief description
G1-G2-G3	Naive / Dense RAG	Retrieval by simple vector similarity
G4-G5-G6	Sparse / BM25 RAG	Lexical retrieval
G7-G8-G9	Hybrid RAG	Dense + sparse fusion
G10–G11–G12	RAG reranking	Initial retrieval + cross-encoder reranker
G13–G14–G15	Multimodal RAG	Image-to-image + image-to-text retrieval
G16-G17-G18	Contextual Compression RAG	Compression of retrieved documents

G19-G20-G21	Self-RAG	Self-evaluation and adaptive retrieval
G22-G23-G24	Corrective RAG	Error detection and correction
G25-G26-G27	RAG query expansion	Query reformulation prior to retrieval
G28–G29–G30	Metadata-Filtered RAG	Retrieval filtered by visual metadata

6. General architecture of the pipeline



7. Permitted components

Component	Examples	Mandatory
VLM / captioning	BLIP-2, GIT, LLaVA, Qwen-VL, Florence-2	Yes

Image encoder	CLIP, SigLIP, DINOv2	Yes
Object detector	YOLO, DETR, GroundingDINO	Optional
Text encoder	Sentence-BERT, CLIP text encoder	Yes
Vector database	FAISS, ChromaDB, LanceDB	Yes
Reranker	Cross-encoder, Cohere Rerank	Depending on the technique
Generative LLM	GPT-2, Flan-T5, Llama-3-8B, Mistral-7B	Yes

Quantised models and small models are encouraged to meet hardware constraints.

8. RAG knowledge base

Each group must build a dataset containing:

• 10,000 images from COCO train2017

• Their reference captions (5 per image)

• Their embeddings (image + text)

• Metadata: detected objects, attributes, relationships .

Required documentation:

• Index type (flat, IVF, HNSW, etc.)

• Embedding dimensions

• Similarity metric (cosine, L2, IP)

• Database size

• Indexing time

9. Hyperparameter tuning

Each group must carry out a basic tuning process and document it:

Hyperparameter	Values tested	Selected value	Justification
top_k retrieval	1, 3, 5, 10	?	?
Chunk size	128, 256, 512 tokens	?	?

Generation temperature	0.1, 0.5, 1.0	?	?
Embedding model	2–3 candidates	?	?
Similarity threshold	0.5, 0.7, 0.9	?	?

Protocol: tuning on validation images, evaluation on the final test images.

Selection method: 10-fold cross-validation on the validation tuning images, selection of the value maximising the average CIDEr.

10. Required experiments

Each group must carry out at least:

10.1 Baseline without RAG

Generate captions without retrieval to serve as a control.

10.2 Pipeline with assigned RAG

Generate captions using the RAG technique.

10.3 Baseline vs RAG comparison

Measure the improvement achieved by RAG.

10.4 Analysis of hallucinations

On a sample of 200 images:

- Object hallucination rate
- Attribute hallucination rate
- Relational hallucination rate

10.5 Uncertainty assessment

- Extract a confidence score for each caption (log-prob, entropy, self-reported confidence).
- Correlate confidence and quality (CIDEr, SPICE).

10.6 Self-assessment and correction (for G13–G16 only)

- Does the system detect its own errors?
- Does correction improve the caption?

10.7 Statistical tests

For each pair of methods being compared (baseline vs RAG, naive RAG vs advanced RAG, etc.):

- Shapiro–Wilk test to check the normality of the score distributions.
- If the distributions are normal: Student's paired t-test ($\alpha = 0.05$).

• If distributions are not normal: Wilcoxon signed-rank test ($\alpha = 0.05$).

• Bonferroni correction for multiple comparisons (corrected $\alpha = 0.05$ / number of comparisons).

• Effect size: Cohen's d for the t-test, Rosenthal's r for the Wilcoxon test.

10.8 Ablation analysis

To identify the contribution of each component of the RAG pipeline:

• Ablation 1: Removal of retrieval (baseline without RAG) ; already covered in 10.1.

• Ablation 2: Varying the number of retrieved evidence items (top_k = 1, 3, 5, 10).

• Ablation 3: Removal of the reranker.

• Ablation 4: Removal of contextual compression.

• Ablation 5: Replacement of the image encoder (CLIP vs SigLIP vs DINOv2).

• Ablation 6: Variation in the generative model (GPT-2 vs Flan-T5 vs Llama-3-8B).

• Ablation 7: Impact of retrieval quality (oracle vs actual).

• Ablation 8: RAG query expansion (G25-G27):

  · Removal of query expansion (original query only).

  · Comparison of variants: simple expansion / multi-query / HyDE.

  · Measurement of the impact on Recall@k, CIDEr, SPICE and hallucination rates.

• Ablation 9: Metadata-Filtered RAG (G28-G30):

  · Removal of filtering/boosting based on visual metadata.

  · Comparison: strict filtering vs. score boosting vs. no metadata.

  · Evaluation of the contribution of detected objects/attributes (YOLO, DETR, GroundingDINO) to accuracy and visual grounding.

Each ablation must be evaluated using the same metrics as the main experiment, and the results compared statistically.

11. Evaluation metrics

11.1 Caption quality

• BLEU-4

• METEOR

• ROUGE-L

• CIDEr (primary metric)

• SPICE

• BERTScore

11.2 Retrieval

• Recall@k (k = 1, 5, 10)

• Precision@k

• MRR (Mean Reciprocal Rank)

• Retrieval latency (ms)

11.3 Hallucination

• Object hallucination rate

• Attribute hallucination rate

• Relation hallucination rate

11.4 Uncertainty

• Calibration (ECE - Expected Calibration Error)

• Confidence/CIDER correlation

11.5 Self-correction (G13-G16)

• Error detection rate

• Improvement rate after correction

11.6 Statistical tests

• p-value (α threshold = 0.05, Bonferroni correction)

• Effect size (Cohen's d, Rosenthal's r)

• 95% confidence interval

• Post-hoc statistical power

12. Qualitative analysis

Each group must provide:

12.1 Examples of success

Image → Visual analysis → Evidence retrieved → Caption generated → Evaluation

12.2 Examples of failures

Image → Visual analysis → Evidence retrieved → Incorrect caption → Hallucination → (Correction or failure)

Each example must be commented on: why the system succeeded or failed.

13. Deliverables (on GitHub)

13.1 Technical report in English (15-20 pages)

Structure:

1. Introduction
2. Problem statement

3. State of the art (RAG + captioning)
4. Assigned RAG technique
5. System architecture
6. Dataset and knowledge base
7. Experimental protocol
8. Quantitative results
9. Statistical tests
10. Ablation analysis
11. Analysis of hallucinations
12. Qualitative analysis
13. Discussion
14. Limitations
15. Conclusion

13.2 Source code

- GitHub repository required
- Complete README.md (installation, usage, reproduction)
- requirements.txt
- Notebooks or modular scripts
- Visible commit history (not a single final commit)

13.3 Dataset of generated captions

CSV/JSON file containing:

- Image ID
- Generated caption
- Method used
- Confidence scores
- Evaluation metrics
- Statistical test results

13.4 Oral presentation in English (15 mins + 15 mins for questions)

- RAG technique
- Architecture
- Results
- Statistical tests and ablation analysis

- Live demonstration
- Limitations and prospects

Presentation schedule:

Phase	**Deadline**
Data exploration and pre-processing	4 November 2026
Validation by the professors	/
Model training and validation	2 December 2026
Model evaluation	21 December 2026

Important:

Any absence from the presentation whether justified or unjustified, will result in a grade 0/20.

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